

REGHURAM KESAVAN

AERODYNAMICS · CFD · AIRCRAFT PERFORMANCE · PYTHON · FLIGHT PHYSICS

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PROFESSIONAL PROFILE

Mechanical Engineering graduate completing the Master in Aerospace Engineering at ISAE-SUPAERO, specialising in Advanced Aerodynamics & Propulsion. Experience spans wind-tunnel/RANS high-lift analysis, Python aircraft-aerodynamics modelling, performance/stability trades, propulsion–airframe interaction and aerodynamic data/surrogate workflows, complemented by aircraft design/test support at Feynman and professional aerospace test-hardware/mechanical-design experience at Lilium.

AERODYNAMICS & FLIGHT PHYSICS

High-Lift Aerodynamics — Wind-Tunnel Testing & Transition-SST RANS ISAE-SUPAERO | 2025

- Conducted low-speed wind-tunnel testing of the HLHLS4 airfoil at $Re = 1 \times 10^6$ across incidence to characterise aerodynamic trends and stall progression.
- Performed Transition-SST RANS/XFOIL analysis using $y^+ < 1$ structured grids (~11k/23k/40k cells); medium-to-fine differences at 10° were ~0.15% CL and ~2% CD, with computed pressure/Cf trends used to assess separation behaviour.

VOROM-X — Python Aerodynamic Design & Flight-Physics Prototype Research Project | 2025 – 2026

- Developed a Python 3D VLM/Trefftz-drag research prototype for sweep/washout, spanwise-loading, efficiency, trim and static-stability trades; applied it to a tailless BWB design study and benchmarked aerodynamic trends against published measurements and XFLR5 to define model limits.

B.Tech Thesis — Passenger Aircraft with Distributed Electric Propulsion APJAKTU | 2021 – 2022

- Compared a clean transport-wing baseline with simplified stationary nacelle installations using steady RANS in ANSYS Fluent, analysing pressure, velocity and downstream boundary-layer/wake changes to assess propulsion–airframe installation effects.

ApexSim — CFD Automation & Aerodynamic Data Methods Independent Project | 2026

- Developed Python workflows for automated SU2 RANS execution, numerical-evidence checks and surrogate/aero-map methods; boundary-condition remediation reduced domain sensitivity from 7.22% → 1.52% in CL and 4.21% → 2.93% in CD.

PROFESSIONAL AEROSPACE EXPERIENCE

Lilium eAircraft GmbH — Mechanical Design Engineer Intern Munich, Germany | Apr – Nov 2024

- Defined compression, tensile and insert-pull-out test requirements and designed/assembled Siemens NX/GD&T test hardware for lightweight sandwich structures; performed pre-test nonlinear FEM in ANSYS Workbench to support test planning and interpretation, with controlled documentation in Teamcenter.

Feynman Aerospace LLP — Aircraft Design Intern Mumbai, India | Apr – Oct 2021

- Performed low-Re aerodynamic/configuration analysis for an ~2 m fixed-wing UAV using XFLR5 and ANSYS Fluent; generated geometry in SolidWorks and supported prototype flight testing and subsequent design iteration.

EDUCATION

ISAE-SUPAERO — Master in Aerospace Engineering Toulouse, France | 2025 – Exp. Dec 2026

Specialisation: Advanced Aerodynamics & Propulsion · Selected: Adv Aero 15/20 · Num Fluid Mech 15/20 · FSI 16/20

Technical University of Munich (TUM) — Graduate Aerospace Engineering Studies Munich, Germany | 2023 – 2025

57 ECTS completed: Aircraft Design · Aircraft Performance · Experimental Aerodynamics · Control Engineering · Aeroelasticity

APJ Abdul Kalam Technological University — B.Tech (Honours), Mechanical Engineering Kerala, India | 2018 – 2022

Official CGPA: 9.28 / 10.0 · Core focus: Fluid Mechanics, Gas Dynamics, Machine Design

TECHNICAL SKILLS

- Aerodynamics & Flight Physics:** RANS CFD · High-Lift · Aircraft Performance · Trim/Static Stability · 3D VLM/Trefftz · Propulsion–Airframe Interaction
- CFD & Data Methods:** ANSYS Fluent · SU2 · XFOIL/XFLR5 · Structured Meshing · y^+ Assessment · Mesh Sensitivity · Aero Maps · Surrogate Modelling
- Programming:** Python (NumPy, SciPy, Matplotlib) · MATLAB · Git/GitHub · Linux/Bash
- Engineering Tools:** Siemens NX · SolidWorks · GD&T · ANSYS Workbench · Teamcenter